Hyperbolas

ANSWER KEY

Standard form

1 For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, find:

a the vertices $(\pm 4, 0)$.

b the foci $c^2 = 16 + 9 = 25$, so foci $(\pm 5, 0)$.

c the equations of the asymptotes $y = \pm \frac{3}{4}x$.

2 Write the standard-form equation of the hyperbola with vertices $(0, \pm 2)$ and foci $(0, \pm 3)$.
$a = 2$, $c = 3$, $b^2 = 9 - 4 = 5$: $\frac{y^2}{4} - \frac{x^2}{5} = 1$.

3 Consider the equation $4x^2 - y^2 + 8x + 4y - 4 = 0$.

a Write the equation in standard form by completing the square. $4(x+1)^2 - (y-2)^2 = 4$, i.e. $(x+1)^2 - \frac{(y-2)^2}{4} = 1$.

b Identify the conic and state its center. A hyperbola opening left-right with center $(-1, 2)$.

Identifying conics

4 Identify each conic section from its equation:

a $x^2 + y^2 = 49$ Circle.

b $\frac{x^2}{9} + \frac{y^2}{25} = 1$ Ellipse.

c $y^2 - 4x^2 = 16$ Hyperbola.

d $y = 3x^2 - 6x + 1$ Parabola.

5 A hyperbola has asymptotes $y = \pm 2x$ and vertices $(\pm 3, 0)$. Find its equation.
$a = 3$ and $\frac{b}{a} = 2$ gives $b = 6$: $\frac{x^2}{9} - \frac{y^2}{36} = 1$.